

Python: Basic to Advanced Short Notes

Introduction

Python is a high-level, interpreted, and easy-to-learn programming language.

Variables & Data Types

int, float, str, bool, complex. Example: `x=10`, `name='Malik'`.

Operators

Arithmetic, Comparison, Logical, Assignment, Bitwise, Membership, Identity.

Conditional Statements

if, elif, else are used for decision making.

Loops

for and while loops are used for repetition. break, continue, pass control loop execution.

Strings

Immutable sequence of characters. Common methods: `upper()`, `lower()`, `split()`, `replace()`.

Lists

Ordered, mutable collection. Example: `[1,2,3]`. Methods: `append()`, `remove()`, `sort()`.

Tuples

Ordered, immutable collection. Example: `(1,2,3)`.

Sets

Unordered collection of unique values. Example: `{1,2,3}`.

Dictionaries

Key-value pairs. Example: `{'name':'Malik','age':20}`. Methods: `keys()`, `values()`, `items()`.

Functions

Reusable block of code. Defined using `def` keyword. Supports arguments and return values.

Lambda Functions

Anonymous one-line functions. Example: `lambda x: x*2`.

Modules & Packages

Modules are Python files. Packages are collections of modules.

Exception Handling

try, except, else, finally used to handle runtime errors.

File Handling

open(), read(), write(), append(), close().

Object-Oriented Programming (OOP)

Classes and Objects, Constructor (`__init__`), Inheritance, Polymorphism, Encapsulation, Abstraction.

Inheritance

Child class acquires properties and methods of parent class.

Polymorphism

Same method behaves differently in different classes.

Encapsulation

Binding data and methods together, restricting direct access.

Abstraction

Hiding implementation details and showing only essential features.

List Comprehension

Compact way to create lists. Example: `[x*x for x in range(5)]`.

Generators

Use yield keyword to generate values one at a time.

Decorators

Functions that modify behavior of another function.

Iterators

Objects implementing `__iter__()` and `__next__()`.

Regular Expressions

re module used for pattern matching and text processing.

SQLite Database

sqlite3 module for local database operations.

MySQL/TiDB Connectivity

Use pymysql or SQLAlchemy for database connections.

Flask Basics

Micro web framework. Routes, templates, forms, and APIs.

API Handling

requests module used to consume APIs. Flask can create APIs.

Virtual Environment

venv isolates project dependencies.

Advanced Topics

Multithreading, Asyncio, Web Scraping, Machine Learning, Data Analysis.